34.4: Tetrapods are gnathostomes that have limbs

One of the most significant events in vertebrate history took place by 365 million years ago, as the fins of a lineage of lobe-fins gradually evolved into the limbs and feet of tetrapods. Until then, all vertebrates had shared the same basic fishlike anatomy. After the colonization of land, tetrapods diversified greatly and gave rise to many new forms, from leaping frogs to flying eagles to bipedal humans.

Derived Characters of Tetrapods

The most significant character of **tetrapods** gives the group its name, which means “four feet” in Greek. In place of pectoral and pelvic fins, tetrapods have limbs with digits. Limbs support a tetrapod’s weight on land, while feet with digits efficiently transmit muscle-generated forces to the ground when it walks.

Life on land selected for numerous other changes to the tetrapod body plan. In tetrapods, the head is separated from the body by a neck that originally had one vertebra on which the skull could move up and down. Later, with the origin of a second vertebra in the neck, the head could also swing from side to side. The bones of the pelvic girdle, to which the hind legs are attached, are fused to the backbone, permitting forces generated by the hind legs against the ground to be transferred to the rest of the body. Except for some fully aquatic species (such as the axolotl; see Figure 42.1), the adults of living tetrapods do not have gills; during embryonic development, the pharyngeal clefts instead give rise to parts of the ears, certain glands, and other structures.

We’ll discuss later how some of these characters were dramatically altered or lost in various lineages of tetrapods. In birds, for example, the pectoral limbs became wings, and in whales, the entire body converged toward a fishlike shape.

The Origin of Tetrapods

As you have read, the Devonian coastal wetlands were home to a wide range of lobe-fins. Those that entered shallow, oxygen-poor water could have used their lungs to breathe air. Some species probably used their stout fins to swim and “walk” underwater across the bottom (moving their fins in an alternating gait, as do some living lobe-fins). This suggests that the tetrapod body plan did not evolve “out of nowhere” but was simply a modification of a preexisting body plan.

The recent discovery of a fossil called *Tiktaalik* provided new details on how this process of modification occurred **(Figure 34.20)**. Like a fish, this species had fins, gills, and lungs, and its body was covered in scales. But unlike a fish, *Tiktaalik* had a full set of ribs that would have helped it breathe air and support its body. Also unlike a fish, *Tiktaalik* had a neck, allowing it to move its head about. In addition, the bones of *Tiktaalik*'s front fin have the same basic pattern found in all limbed animals: one bone (the humerus), followed by two bones (the radius and ulna), followed by a group of small bones that comprise the wrist. Finally, *Tiktaalik*'s pelvis and rear fin were larger and more robust than those of a fish; the pelvis is the bony structure to which hind limbs are attached in tetrapods. Although it is unlikely that *Tiktaalik* could walk on land, the skeletal structure of its fins and pelvis suggests that it could prop itself up and walk in water on its fins. Since *Tiktaalik* predates the oldest known tetrapod, its features suggest that key "tetrapod" traits, such as a wrist, ribs, and a neck, were in fact ancestral to the tetrapod lineage.

Figure 34.20

Discovery of a "fishapod": *Tiktaalik*.

Tiktaalik fossil has flat, roughly triangular head with blunt end narrowing at neck, and parts of ribs and fin visible.

The fish and tetrapod are compared as follows:

Fish Characters	Tetrapod Characters	Other Structures Noted in Image of fishapod
Scales	Neck	Head
Fins	Ribs	Shoulder bones (2)
Gills and lungs	Fin skeleton with elbow, radius, humerus, ulna and wrist	blank
blank	Flat skull	blank
blank	Eyes on top of skull (2)	blank

Paleontologists were on the hunt for fossils that could shed light on the evolutionary origin of tetrapods. Based on the ages of previously discovered fossils, researchers were looking for a dig site with rocks about 365–385 million years old. Ellesmere Island, in the Canadian Arctic, was one of the few such sites that was also likely to contain fossils, because it was once a river. The search at this site was rewarded by the discovery of fossils of a 375-million-year-old lobe-fin, named *Tiktaalik*. As shown in the chart and photographs, *Tiktaalik* exhibits both fish and tetrapod characters.

MAKE CONNECTIONS *Describe how* Tiktaalik's *features illustrate Darwin's concept of descent with modification (see*Concept 22.2*).**

*For suggested answer, see*Appendix A.**

HHMI Video

Great Transitions: The Origin of Tetrapods

image

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Tiktaalik and other extraordinary fossil discoveries have allowed paleontologists to reconstruct how fins became progressively more limb-like over time, culminating in the appearance in the fossil record of the first tetrapods 365 million years ago **(Figure 34.21)**. Over the next 60 million years, a great diversity of tetrapods arose. Some of these species retained functional gills and had weak limbs, while others lost their gills and had stronger limbs that facilitated walking on land. Overall, judging from the morphology and locations of their fossils, most of these early tetrapods probably remained tied to water, a characteristic they share with some members of the most basal group of living tetrapods, the amphibians.

Figure 34.21

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Steps in the origin of limbs with digits.
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An evolutionary tree diagram.

The white bars on the branches of this diagram place known fossils in time; arrowheads indicate lineages that extend to today. The drawings of extinct organisms are based on fossilized skeletons, but the colors are fanciful.

The tree is represented by the following list.

Paleozoic: Branch point 1; Silurian; a little over 415 million years ago

Lungfishes; about 410 million years ago persisting to present; humerus only

Branch point 2; Devonian; 400 million years ago

Eusthenopteron (fish) from 385 to about 380 million years ago; thick humerus, long radius, shorter thick ulna

Branch point 3; Devonian; about 390 million years ago

Panderichthys (fish) from a little over 385 to a little under 385 million years ago; nearly rectangular humerus, thick radius and ulna

Branch point 4; Devonian; a little over 385 million years ago

Tiktaalik (fishapod); Devonian; from a little over 370 to a little under 370 million years ago; wide humerus, long, thick radius, short thick ulna

(Limbs with digits); Branch point 5; Devonian; about 370 million years ago

Acanthostega; (amphibian-like); Devonian; persisted for a few million years a little under 370 million years ago; wide humerus, thin radius and ulna, digits

Branch point 6; Devonian; about 370 million years ago

Tulerpeton; (amphibian like with longer limbs) Devonian; persisted a few million years about 360 million years ago; thick L-shaped humerus, thin radius and ulna, digits

Branch point 7; Devonian; just under 370 million years ago

Amphibians; Carboniferous; from 340 million years ago to present

Amniotes; Carboniferous; from about 320 million years ago to present

VISUAL SKILLS *Assuming the branch point dates shown in this diagram are accurate, what range of dates would include the origin of amphibians?*

*For suggested answer, see*Appendix A*.*

Amphibians

An evolutionary tree highlights amphibia.

Branch point 1: First level-1 branch: Cephalochordata. The second level-1 branch splits from branch point 2 into the first level-2 branch: Urochordata. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Myxini and the second level-4 branch: Petromyzontida. The second level-3 branch splits from branch point 5 into the first level-5 branch: Chondrichthyes. The second level-5 branch splits from branch point 6 into the first level-6 branch: Actinopterygii. The second level-6 branch splits from branch point 7 into the first level-7 branch: Actinista. The second level-7 branch splits from branch point 8 into the first level-8 branch: Dipnoi. The second level-8 branch splits from branch point 9 into the first level-9 branch: Amphibia. The second level-9 branch splits from branch point 10 into the first level-10 branch: Reptilia and the second level-10 branch: Mammalia.

The **amphibians** are represented today by about 6,150 species in three clades: salamanders (clade Urodela, “tailed ones”), frogs (clade Anura, “tailless ones”), and caecilians (clade Apoda, “legless ones”).

Salamanders

There are about 550 known species of urodeles, or salamanders. Some are entirely aquatic, but others live on land as adults or throughout life. Most salamanders that live on land walk with a side-to-side bending of the body, a trait also found in early terrestrial tetrapods **(Figure 34.22a)**. Paedomorphosis is common among aquatic salamanders; the axolotl, for instance, retains larval features even when sexually mature (see Figure 25.25).

Figure 34.22

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Amphibians.
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Three amphibians.

Part a. Order Urodela. Urodeles (salamanders) retain their tail as adults.

Part b. Order Anura. Anurans (toads and frogs) lack a tail as adults.

Part c. Order Apoda. Apodans, or caecilians, are legless, mainly burrowing amphibians.

Frogs

Numbering about 5,420 species, anurans, or frogs, are better suited than salamanders to locomotion on land **(Figure 34.22b)**. Adult frogs use their powerful hind legs to hop along the terrain. Although often distinctive in appearance, the animals known as “toads” are simply frogs that have leathery skin or other adaptations for life on land. A frog nabs insects and other prey by flicking out its long, sticky tongue, which is attached to the front of the mouth. Frogs display a great variety of adaptations that help them avoid being eaten by larger predators. Their skin glands secrete distasteful or even poisonous mucus. Many poisonous species have color patterns that camouflage them or have bright coloration, which predators appear to associate with danger (see Figure 54.6).

Caecilians

The approximately 170 species of apodans, or caecilians, are legless and nearly blind, and superficially they resemble earthworms **(Figure 34.22c)**. Their absence of legs is a secondary adaptation, as they evolved from a legged ancestor. Caecilians inhabit tropical areas, where most species burrow in moist forest soil.

Lifestyle and Ecology of Amphibians

The term *amphibian* (derived from *amphibious*, meaning “both ways of life”) refers to the life stages of many frog species that live first in water and then on land **(Figure 34.23)**. The larval stage of a frog, called a tadpole, is usually an aquatic herbivore with gills, a lateral line system resembling that of aquatic vertebrates, and a long, finned tail. The tadpole initially lacks legs; it swims by undulating its tail. During the metamorphosis that leads to the “second life,” the tadpole develops legs, lungs, a pair of external eardrums, and a digestive system adapted to a carnivorous diet. At the same time, the gills disappear; the lateral line system also disappears in most species. The young frog crawls onto shore and becomes a terrestrial hunter. In spite of their name, however, many amphibians do not live a dual—aquatic and terrestrial—life. There are some strictly aquatic or strictly terrestrial frogs, salamanders, and caecilians. Moreover, salamander and caecilian larvae look much like the adults, and typically both the larvae and the adults are carnivorous.

Figure 34.23

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The “dual life” of a frog (*Rana temporaria*).
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The dual life of a frog.

Part a. The tadpole is an aquatic herbivore with a fishlike tail and internal gills.

Part b. During metamorphosis, the gills and tail are resorbed, and walking legs develop. The adult frog will live on land.

Part c. The adults return to water to mate. The male grasps the female, stimulating her to release eggs. The eggs are laid and fertilized in water. They have a jelly coat but lack a shell and would desiccate in air.

Most amphibians are found in damp habitats such as swamps and rain forests. Even those adapted to drier habitats spend much of their time in burrows or under moist leaves, where humidity is high. One reason amphibians require relatively wet habitats is that they rely heavily on their moist skin for gas exchange—if their skin dries out, they cannot get enough oxygen. In addition, amphibians typically lay their eggs in water or in moist environments on land; their eggs lack a shell and dehydrate quickly in dry air.

Fertilization is external in most amphibians; the male grasps the female and spills his sperm over the eggs as the female sheds them (see Figure 34.23c). Some amphibian species lay vast numbers of eggs in temporary pools, and egg mortality is high. In contrast, other species lay relatively few eggs and display various types of parental care. Depending on the species, either males or females may house eggs on their back **(Figure 34.24)**, in their mouth, or even in their stomach. Certain tropical tree frogs stir their egg masses into moist, foamy nests that resist drying.

Figure 34.24

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A mobile nursery.
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A female pygmy marsupial frog on a leaf has at least six large, rounded bulges (eggs) on her back beneath the skin.

A female marsupial frog (Flectonotus fitzgeraldi) incubates her eggs in pouches of skin on her back.

Many amphibians exhibit complex and diverse social behaviors, especially during breeding seasons. Frogs are usually quiet, but the males of many species vocalize to defend their breeding territory or to attract females. In some species, migrations to specific breeding sites may involve vocal communication, celestial navigation, or chemical signaling.

Over the past 30 years, zoologists have documented a rapid and alarming decline in amphibian populations in locations throughout the world. There appear to be several causes, including the spread of a disease-causing chytrid fungus (see Figure 31.25), habitat loss, climate change, and pollution. In some cases, declines have become extinctions. Recent studies indicate that at least nine amphibian species have become extinct within the last four decades; more than 100 other species have not been observed in that time and are considered possibly extinct. In the **Problem-Solving Exercise**, you can explore one possible strategy to prevent amphibian deaths from fungal infections.

Problem-Solving Exercise

AP

Can declining amphibian populations be saved by a vaccine?

Amphibian populations are declining rapidly worldwide. The fungus *Batrachochytrium dendrobatidis* (Bd) has contributed to this decline: This pathogen causes severe skin infections in many amphibian species, leading to massive die-offs. Efforts to save amphibians from Bd have had limited success, and there is little evidence that frogs and other amphibians have acquired resistance to Bd on their own.

Can declining amphibian populations be saved by a vaccine?

Yellow-legged frogs (Rana muscosa) in California killed by Bd infection

Amphibian populations are declining rapidly worldwide. The fungus Batrachochytrium dendrobatidis (B d) has contributed to this decline: This pathogen causes severe skin infections in many amphibian species,

leading to massive die-offs. Efforts to save amphibians from B d have had limited success, and there is little evidence that frogs and other amphibians have acquired resistance to B d on their own. In a photograph, yellow-legged frogs (Rana muscosa) in California are belly up in water, having been killed by a B d infection.

In this exercise, you will investigate whether amphibians can acquire resistance to the fungal pathogen Bd.

Your Approach The principle guiding your investigation is that prior exposure to a pathogen can enable amphibians to acquire immunological resistance to that pathogen. To see whether this occurs after exposure to Bd, you will analyze data on acquired resistance in Cuban tree frogs (*Osteopilus septentrionalis*).

Your Data To create variation in number of prior exposures to Bd, Cuban tree frogs were exposed to Bd and cleared of their infection (using heat treatments) from zero to three times; frogs with no prior exposures are referred to as "naive." Researchers then exposed frogs to Bd and measured mean abundance of Bd on the frog's skin, frog survival, and abundance of lymphocytes (a type of white blood cell involved in the vertebrate immune response).

Two graphs and one table. The number of B d organisms on the frogs' skin fell with the number of prior B d exposures. Percentage survival peaked with two prior B d exposures.

Two graphs and one table are as follows.

Graph on left: Y-axis shows B d abundance frog's skin (organisms/g of frog) from 12,000 to 24,000, marked in increments of 2000. X-axis shows number of prior Rd exposures from zero to 4 in increments of one. See Problem-Solving Table A.

Middle graph: Y-axis shows percentage survival from zero to 0.8, marked in increments of 0.1. X-axis shows number of prior B d exposures from zero to four in increments of one. See Problem-Solving Table B.

Table on right compares number of prior B d exposure with thousands of lymphocytes per gram frog. Zero exposures = 134,000 lymphocytes. 1 exposure = 240,000 lymphocytes. 2 exposures = 244,000 lymphocytes. 3 exposures = 227,000 lymphocytes.

A table illustrates the number of prior B d exposures and thousands of lymphocytes per gram of frog.

Number of prior B d exposures	Thousands of lymphocytes per gram of frog
0	134
1	240
2	244

3	227

Your Analysis

- Describe and interpret the results shown in the figure.
-
  - Graph the data in the table.
- Based on these data, develop a hypothesis that explains the results discussed in question 1.
- Breeding populations of amphibian species threatened by Bd have been established in captivity. In addition, evidence suggests that Cuban tree frogs can acquire resistance after exposure to dead Bd. Based on this information and your answers to questions 1 and 2, suggest a strategy for repopulating regions decimated by Bd.

Instructors: A version of this Problem-Solving Exercise can be assigned in **Mastering Biology**.